Chapter 12: Arithmetic Signatures of the Figure-eight Knot

Chapter 11 identified a candidate transform space for the minimal orientable layer, and a conjectural linkage spectrum between such layers. We now return to the figure-eight knot itself.

If the figure-eight knot complement supplies the internal geometry of persistence, then its arithmetic should not be incidental. It should leave signatures: privileged phases, special functions, volume constants, endpoint values, and descendant channels that can be compared against the constructive alphabet of the Transform Dictionary.

The question of this chapter is: what arithmetic is generated by the figure-eight knot volume?

The hyperbolic figure-eight knot volume is expressed by a conjugate dilogarithm difference evaluated at the imaginary golden ratio and its inverse.[77]

$$\mathrm{V_{fe}} = i\left[Li_2\left(\frac{1}{\varphi_i}\right) - Li_2(\varphi_i)\right]$$

This construction uses the imaginary unit $i = (-1)^{1/2}$ multiplied by the difference between two dilogarithms $Li_2(x)$ evaluated at the inverse imaginary golden ratio $1/\varphi_i = (-1)^{-1/3}$ and the imaginary golden ratio $\varphi_i = (-1)^{1/3}$ (principal branch).

These unit-phases form the minimal phase seed of the figure-eight knot's dilogarithmic construction. Everything that follows—the special functions, constants, endpoint values, and arithmetic companions that appear throughout the Transform Dictionary—can be read as unfolding from this unit-phase seed.

Each dilogarithm in the knot's construction decomposes into a real part and an imaginary part. The real parts match and cancel in the volume difference; the imaginary parts oppose each other and combine to produce the hyperbolic volume.

$$Li_2\left(\frac{1}{\varphi_i}\right) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 - G_{Gi}\, i \qquad Li_2(\varphi_i) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 + G_{Gi}\, i$$

where $\Gamma(x)$ is the gamma function, $\Gamma(5) = 24$, and G_{Gi} is Gieseking's constant. This decomposition exposes three irreducible components of the knot—a real modular partition scale $\left(\frac{4\pi}{\Gamma(5)}\right)^2$, an imaginary hyperbolic volume scale G_{Gi}, and the functional generator itself, the dilogarithm $Li_2(x)$.

The dilogarithm[78] defines the power series expansion,

$$Li_2(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2} = \frac{x^1}{1^2} + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \frac{x^4}{4^2} + \frac{x^5}{5^2} + \cdots.$$

Its derivative is controlled by the natural logarithm,

$$\frac{d}{dx}\big(Li_2(x)\big) = \frac{Li_1(x)}{x} = -\frac{\log(1-x)}{x}.$$

On the real unit interval, $Li_2(x)$ increases to its endpoint value

$$Li_2(1) = \zeta(2).$$

Taking the logarithmic integral of $Li_2(x)$ yields the trilogarithm $Li_3(x)$ whose endpoint value is $\zeta(3)$,

$$Li_3(1) = \zeta(3).$$

The endpoint relation

$$Li_s(1) = \zeta(s)$$

connects the polylogarithmic ladder to the zeta ladder, initially for $Re(s) > 1$ and then by analytic continuation.

Thus, embedded directly within the arithmetic of the figure-eight knot volume V_{fe}, we find the geometric constellation that organizes the constants of Nature in the Transform Dictionary: the dilogarithm $Li_2(x)$, the natural logarithm $\log(x)$, the gamma function $\Gamma(x)$, the Riemann zeta function $\zeta(x)$, Archimedes' constant π, and Gieseking's constant G_{Gi}. These characterize structural properties of the knot.

The same inversion-dilogarithm has a second distinguished unit-circle evaluation at the square phase i. At this phase, the engine shifts from producing twice Gieseking's constant, $2G_{\mathrm{Gi}} = \mathrm{V_{fe}}$, to producing twice Catalan's constant, $2K$.[79]

$$2K = i\left[Li_2\left(\frac{1}{i}\right) - Li_2(\,i\,)\right]$$

The same core dilogarithmic structure generates both the hyperbolic volume of the figure-eight knot and Catalan's constant—the alternating sum of odd reciprocals squared.[80]

Thus, the arithmetic backbone of the hyperbolic figure-eight knot volume singles out a minimal generating arithmetic set:

unit phases: $1, i, \varphi_i, {\varphi_i}^{-1}$

functions: $Li_3(x), Li_2(x), \log(x), \Gamma(x), \zeta(x)$

constants: $\mathrm{V_{fe}}, G_{Gi}, 2K, 4\pi, \zeta(2), \zeta(3), \Gamma(5)$

This list separates naturally into geometric habitats. The phases 1, i, φ_i, and ${\varphi_i}^{-1}$ belong to the unit circle: they are phase addresses, roots and rotations that determine how the dilogarithm is sampled. Catalan's constant K, Archimedes' constant π, and the trigonometric relations that follow are circular descendants of this unit-phase structure. The lemniscatic constants s, L, L_1, L_2, G_{Ga}, and C_U, together with the elliptic periods ω_1 and ω_2, belong to the unit lemniscate: the elliptic companion to the unit circle, governed by gamma-function values such as $\Gamma^2\left(\frac{1}{4}\right)$ and $\Gamma^3\left(\frac{1}{3}\right)$. The hyperbolic constants $\mathrm{V_{fe}}$ and G_{Gi} belong to the figure-eight knot itself, where the unit-circle phases are lifted into hyperbolic volume through the dilogarithm.

Algebraic and Metric Combinations

The first descendants of this seed appear through algebraic and metric combination. This includes Euler's number e,[81] the domino tiling constant C_d,[82] the dimer constant D_d,[83] the arc length of the unit lemniscate s,[84] the lemniscate constant L, the 1st lemniscate constant L_1, the 2nd lemniscate constant L_2, Gauss's constant G_{Ga},[85] the ubiquitous constant

C_U,[86] and Sierpiński's constant S.[87] The universal parabolic constant P_{up}[88] enters as the parabolic companion to the circular factor $4\pi/8$.

$$\left(i^i\right)^2 = e^{-\pi} \qquad \log\left(i^i\right) = -\frac{4\pi}{8} \qquad C_d = \frac{K}{\pi} \qquad D_d = e^{2C_d}$$

$$s = 2L \qquad L = 2L_1 \qquad L_1 = \frac{s}{4} \qquad L_2 = \frac{\pi}{s} \qquad G_{Ga} = \frac{s}{2\pi} \qquad C_U = \sqrt{2}\,L_2$$

$$S = \log\left(\left(e^{\gamma}\,C_U\right)^2\right) \qquad P_{up} = \int_{-1}^{1} \sqrt{1+x^2} \qquad \frac{4\pi}{8} = \int_{-1}^{1} \sqrt{1-x^2}$$

Gamma Function Constants

The gamma-function branch enters next. The lemniscate constants, the elliptic periods ω_1 and ω_2,[89] and the Euler-Mascheroni constant γ[90] are also organized by gamma-function evaluations and derivatives. The unit lemniscate appears as the elliptic companion to the unit circle, while γ enters through the logarithmic derivative of $\Gamma(x)$.

$$s = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{2\pi}} \qquad L = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{8\pi}} \qquad L_1 = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{32\pi}}$$

$$L_2 = \frac{2\pi}{\sqrt{2}}\,\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)} \qquad C_U = 2\pi\,\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)} \qquad G_{Ga} = \frac{\Gamma^2\left(\frac{1}{4}\right)}{2\pi\sqrt{2\pi}}$$

$$\omega_1 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{2\pi}\varphi_i \qquad \omega_2 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{2\pi} \qquad -\gamma = \Gamma'(1)$$

This alphabet is stratified into circular phases, lemniscatic arcs, hyperbolic volumes, parabolic measures, and analytic continuations—different projections of one constructive geometry. This chapter has identified the arithmetic seed, together with its first algebraic, metric, and gamma-function descendants. Chapter 13 follows the remaining channels into the full geometric alphabet of the Transform Dictionary: fixed points, recursive constructions, continued fractions, statistical limits, oscillatory nodes, prime-sensitive products, factorial orderings, derangement exclusions, sparse digit constructions, lattice sums, and asymptotic limits.